Numerical analysis of superconducting phases in the extended Hubbard model with non-local pairing

University of Pisa, a.y. 2025-2026

Alessandro Gori*

Thesis for the Master's degree in Physics

Abstract

[To be continued...]

- Appendix on s-wave SC ghost simulations on HM
- Appendix on triplet pairing in MFT self-consistent scheme

Contents

1 The normal phase **1**
1.1 The "normal" phase and its symmetries . . . 1
1.2 Hartree effects: chemical potential renormalization . . . 1
1.2.1 Hartree channel in the normal phase: analytic derivation . . . 2
1.2.2 Hartree contraction energy . . . 3
1.3 Fock effects: bare bands renormalization . . . 3
1.3.1 Fock channel in the normal phase: analytic derivation . . . 4
1.3.2 Fock contraction energy . . . 6
1.3.3 Nematicity: extending hopping to d-wave bands . . . 6
1.4 Normal free energy density . . . 7
1.5 Theoretical constraints on the normal phase HFPs . . . 9
1.5.1 Constraints in presence of a robust C_4 symmetry and bands shrinking . . . 9
1.5.2 Extension to a $C_2/\mathbb{Z}_2$ symmetry . . . 10
1.6 Numerical simulations . . . 10
1.6.1 HF iterative algorithm sketch . . . 10

Bibliography **13**

Draft: February 27, 2026

*a.gori23@studenti.unipi.it / nepero27178@github.com

List of symbols and abbreviations

AF	Antiferromagnetic
BCS	Bardeen-Cooper-Schrieffer (theory)
β	Inverse temperature
$\Delta^{(\gamma)}$	γ-wave component of the gap function
DoF	Degree(s) of freedom
EHM	Extended Hubbard model
EV(s)	Expectation value(s)
FS	Fermi surface
HF	Hartree-Fock
HFP(s)	Hartree-Fock parameter(s)
HM	Hubbard model
L_ℓ	Number of lattice sites in direction ℓ
LRT	Linear Response Theory
MFT	Mean-Field Theory
NN(s)	Nearest neighbor(s)
RWC(s)	Relevant Wick's contraction(s)
RMP(s)	Renormalized model parameter(s)
SC	Superconductor
SDW	Spin-density wave
SSB	Spontaneous symmetry breaking
T_c	Critical temperature

Chapter 1

The normal phase

This chapter is devoted to the MFT analysis of the "normal phase", which is, the one where interacting electrons do not develop an antiferromagnetic or superconducting gap, although undergoing an effective non-trivial transformation. What will be derived here, in particular the hopping renormalization effect, constitutes a peculiarity of effective MFT treatment of the EHM and will be the groundwork of following analyses. Note that this chapter is structured in such a way that what is derived here mathematically is recovered and expanded to more complex phases in next chapters.

1.1 The "normal" phase and its symmetries

The normal phase is simply the one where no symmetry is broken, and the many-body ground state preserves the entire U(2) structure of Eq. (**??**) as well as full translational symmetry $\mathrm{T}_1^{(x)} \otimes \mathrm{T}_1^{(y)}$. In terms of spatial symmetries, we here theoretically allow for

$$\mathrm{C}_4 \overset{\mathrm{SSB}}{\to} \mathrm{C}_2$$

symmetry reduction and extract the mathematical interesting consequences. While this implies immediately inversion symmetry preservation, full C_4 symmetry will be imposed later.

The normal phase is essentially given by the null gap limit of both the AF and SC phases. By discussing it separately now, we highlight a MFT feature of the EHM of particular interest, namely, the insurgence of a new, self-consistent hopping parameter. From Tabs. **??** and **??**, we know that the relevant Wick's contractions (RWCs) for this phase are

$\hat{H}_U$	Hartree contraction
$\hat{H}_V$ (o.s. sector)	Hartree contraction
$\hat{H}_V$ (s.s. sector)	Hartree and Fock contractions

We anticipate the final effects of all RWCs. As we will see in detail and is reported in Tab. 1.1, the Hartree contractions account essentially for chemical potential shifts, while the only Fock contraction for a renormalization effects on the bare bands.

In the normal phase, spin is a pure physical degeneracy, being the $\uparrow$ and $\downarrow$ sectors separated; moreover, $\hat{H}$ is $\mathbb{Z}_2$ invariant with respect to spin inversion, $\sigma \leftrightarrow \overline{\sigma}$. This implies that all physical quantities, as is for the normal phase, should not depend on the spin index. We will perform all computations leaving σ explicit and using this consideration at the end.

1.2 Hartree effects: chemical potential renormalization

The first RMP we discuss is μ, whose numerical value is renormalized by Hartree channel interactions within MFT. As is discussed in Sec. **??**, in the context of numerical simulations at fixed density, the chemical potential is determined self-consistently; thus any net effect that shifts μ is effectively ignored within (and limitedly to) the iterative algorithm. Renormalization effects become relevant when computing free energy.

Operator	Sector	RWCs	Net effect
$\hat{H}_U$		Hartree	μ shift by $-nU$
$\hat{H}_V$	o.s.	Hartree	μ shift by nzV
	s.s.	Hartree	μ shift by nzV
		Fock	Bands renormalization and resymmetrization

Table 1.1 | Relevant Wick's contractions and net effect in the normal phase.

1.2.1 Hartree channel in the normal phase: analytic derivation

Within this section we discuss analytically the effects of MFT on the Hartree channel, separating the local and non-local interactions and obtaining a set of HFPs needed to setup an interative HF algorithm.

Local repulsion. In the normal phase, the local repulsion $\hat{H}_U$ decomposes in Hartree channel as

$$\hat{H}_U \simeq U \sum_i \left[\langle \hat{n}_{i\uparrow}\rangle \hat{n}_{i\downarrow} + \hat{n}_{i\uparrow}\langle \hat{n}_{i\downarrow}\rangle - \langle \hat{n}_{i\downarrow}\rangle\langle \hat{n}_{i\uparrow}\rangle\right]$$

For a translationally invariant phase such as the normal phase, it holds

$$\langle \hat{n}_{i\sigma}\rangle = n \tag{1.1}$$

which gives immediately

$$\begin{aligned}\hat{H}_U &\simeq nU \sum_i \left[\hat{n}_{i\uparrow} + \hat{n}_{i\downarrow}\right] - U\sum_i n^2 \\ &= nU \times \hat{N} - E^{(\mathrm{N})}_{\mathrm{H}/U}\end{aligned}$$

being $\hat{N}$ the total number operator and $E^{(\mathrm{N})}_{\mathrm{H}/U}$ the "contraction energy" (also known as double counting term) due to the Hartree (H) contraction of the U repulsion, discussed in Sec. 1.2.2. The chemical potential is corrected by

$$\mu \to \mu - nU \tag{1.2}$$

The non-local attraction further corrects μ.

Non-local attraction. Let us break down $\hat{H}_V$ isolating its Hartree RWCs in both o.s. and s.s. sectors:

$$\begin{aligned}\hat{H}_V \simeq\ & \overbrace{-V\sum_{\langle ij\rangle}\sum_\sigma \left[\langle \hat{n}_{i\sigma}\rangle \hat{n}_{j\sigma} + \hat{n}_{i\sigma}\langle \hat{n}_{j\sigma}\rangle - \langle \hat{n}_{i\sigma}\rangle\langle \hat{n}_{j\sigma}\rangle\right]}^{\text{s.s.}} \\ & \underbrace{-V\sum_{\langle ij\rangle}\sum_\sigma \left[\langle \hat{n}_{i\sigma}\rangle \hat{n}_{j\bar{\sigma}} + \hat{n}_{i\sigma}\langle \hat{n}_{j\bar{\sigma}}\rangle - \langle \hat{n}_{i\sigma}\rangle\langle \hat{n}_{j\bar{\sigma}}\rangle\right]}_{\text{o.s.}} + (\text{all the rest})\end{aligned} \tag{1.3}$$

where "all the rest" collects all non-Hartree RWCs. Recalling the Ansatz of Eq. (1.1), we get

$$\hat{H}_V \simeq \underbrace{-nV\sum_{\langle ij\rangle}\sum_\sigma \left[\hat{n}_{j\sigma} + \hat{n}_{i\sigma}\right]}_{\text{s.s.}} \underbrace{-nV\sum_{\langle ij\rangle}\sum_\sigma \left[\hat{n}_{j\bar{\sigma}} + \hat{n}_{i\sigma}\right]}_{\text{o.s.}} - E^{(\mathrm{N})}_{\mathrm{H}/V} + (\text{all the rest})$$

with $E^{(\mathrm{N})}_{\mathrm{H}/V}$ the normal state shift to energy brought by $\hat{H}_V$. Now, evidently both sums above are just a number operator, thus including the degeneracies introduced by the summations on NNs for a square lattice with coordination number $z = 4$

$$\hat{H}_V \simeq -2nzV \times \hat{N} - E^{(\mathrm{N})}_{\mathrm{H}/V} + (\text{all the rest})$$

This accounts for the final Hartree shift of μ,

$$\mu \to \mu + 2znV \tag{1.4}$$

Considering the net shift to μ due to both interactions by combining Eqns. (1.2), (1.4), we get the final RMP anticipated in Tab. 1.1,

$$\tilde{\mu} \equiv \mu + n(2zV - U) \tag{1.5}$$

This result remains valid throughout all phases discussed in this text: for the AF phase, the SDW character leaves this shift untouched, while the SC phase we will discuss is inherently translational invariant.

1.2.2 Hartree contraction energy

The double counting terms accounting for energy shifts in the Hartree channel are, as anticipated:

$$\begin{aligned} E^{(\mathrm{N})}_{\mathrm{H}/U} &= \sum_i \langle \hat{n}_{i\downarrow}\rangle\langle \hat{n}_{i\uparrow}\rangle \\ &= L_x L_y \times n^2 U \end{aligned} \tag{1.6}$$

for the local repulsion and

$$\begin{aligned} E^{(\mathrm{N})}_{\mathrm{H}/V} &= -V \sum_{\langle ij\rangle}\sum_{\sigma} \left[\langle \hat{n}_{i\sigma}\rangle\langle \hat{n}_{j\sigma}\rangle + \langle \hat{n}_{i\sigma}\rangle\langle \hat{n}_{j\overline{\sigma}}\rangle\right] \\ &= -2V \sum_{\langle ij\rangle}\sum_{\sigma} n^2 \\ &= -L_x L_y \times 2zn^2 V \end{aligned} \tag{1.7}$$

for the non-local attraction. In the last passage, we used Eq. (**??**), as well as spin degeneracy. Note that, correctly, the contraction energy is an extensive quantity. This implies that we can define a contraction energy per site – something we will do later when computing total free energy density.

1.3 Fock effects: bare bands renormalization

Up to now, no relevant changes have been made to the starting situation of the conventional degenerate Fermi gas. The scenario changes now, with the analysis of the Fock RWCs. Even if no gap opens, in the normal phase particles interact by the means of the non-local attraction $\hat{H}_V$, a feature that effectively redefines the bare fermionic bands.

From Wick's decomposition of $\hat{H}_V$, the only allowed Fock term comes from the same-spin part due to SU(2) symmetry selection rules. Let us focus only on this term when decomposing $\hat{H}$:

$$\hat{H}_V \simeq (\text{Hartree channel}) + V \sum_{\langle ij\rangle}\sum_{\sigma} \left[\langle \hat{c}^\dagger_{i\sigma}\hat{c}_{j\sigma}\rangle \hat{c}^\dagger_{j\sigma}\hat{c}_{i\sigma} + \hat{c}^\dagger_{i\sigma}\hat{c}_{j\sigma}\langle \hat{c}^\dagger_{j\sigma}\hat{c}_{i\sigma}\rangle - \langle \hat{c}^\dagger_{i\sigma}\hat{c}_{j\sigma}\rangle\langle \hat{c}^\dagger_{j\sigma}\hat{c}_{i\sigma}\rangle\right] \tag{1.8}$$

(note the + sign in front of the displayed term). A bond-wise hopping amplitude can be defined,

$$\tilde{t}_{ij\sigma} \equiv t - V\langle \hat{c}^\dagger_{j\sigma}\hat{c}_{i\sigma}\rangle$$

The effective kinetic hamiltonian is given by

$$\begin{aligned} \hat{H}_{\tilde{t}} &= \hat{H}_t + V \sum_{\langle ij\rangle}\sum_{\sigma} \left[\langle \hat{c}^\dagger_{i\sigma}\hat{c}_{j\sigma}\rangle \hat{c}^\dagger_{j\sigma}\hat{c}_{i\sigma} + \text{h.c.}\right] \\ &= -\sum_{\langle ij\rangle}\sum_{\sigma} \left[\tilde{t}_{ij\sigma}\hat{c}^\dagger_{i\sigma}\hat{c}_{j\sigma} + \text{h.c.}\right] \end{aligned}$$

To proceed, it is useful to move to reciprocal space.

1.3.1 Fock channel in the normal phase: analytic derivation

In reciprocal space, the effective hopping must be transformed as well. Consider the Fourier Transform $\mathcal{F}$ given in Eq. (**??**), applied to the displayed part of Eq. (1.8),

$$V\sum_{\langle ij\rangle}\sum_{\sigma}\left[\langle\hat{c}^\dagger_{i\sigma}\hat{c}_{j\sigma}\rangle\hat{c}^\dagger_{j\sigma}\hat{c}_{i\sigma}+\hat{c}^\dagger_{i\sigma}\hat{c}_{j\sigma}\langle\hat{c}^\dagger_{j\sigma}\hat{c}_{i\sigma}\rangle-\langle\hat{c}^\dagger_{i\sigma}\hat{c}_{j\sigma}\rangle\langle\hat{c}^\dagger_{j\sigma}\hat{c}_{i\sigma}\rangle\right]$$
$$\overset{\mathcal{F}}{=}\frac{2V}{L_xL_y}\sum_{\mathbf{K},\mathbf{k},\mathbf{k}'}\sum_{\sigma}\left[\cos(\delta k_x)+\cos(\delta k_y)\right]\langle\hat{c}^\dagger_{\mathbf{K}+\mathbf{k}\sigma}\hat{c}_{\mathbf{K}-\mathbf{k}'\sigma}\rangle\hat{c}^\dagger_{\mathbf{K}-\mathbf{k}\sigma}\hat{c}_{\mathbf{K}+\mathbf{k}'\sigma}-E^{(\mathrm{N})}_{\mathrm{F}/V} \tag{1.9}$$

Here, the 2 prefactor comes from recognizing that the first two operators in square brackets in the first line generate identical contributions to the full sum (as is easy to see by performing trivial variables changes); the double counting energy shift is given by

$$E^{(\mathrm{N})}_{\mathrm{F}/V}\equiv\frac{V}{L_xL_y}\sum_{\mathbf{K},\mathbf{k},\mathbf{k}'}\sum_{\sigma}\left[\cos(\delta k_x)+\cos(\delta k_y)\right]\langle\hat{c}^\dagger_{\mathbf{K}+\mathbf{k}\sigma}\hat{c}_{\mathbf{K}-\mathbf{k}'\sigma}\rangle\langle\hat{c}^\dagger_{\mathbf{K}-\mathbf{k}\sigma}\hat{c}_{\mathbf{K}+\mathbf{k}'\sigma}\rangle \tag{1.10}$$

Up to this point, this derivation is general and will be re-used in next chapters. Here the specificity of the physical phase settles in: if we assume the normal state to be a free degenerate Fermi gas with renormalized bands, we get

$$\langle\hat{c}^\dagger_{\mathbf{k}_1\sigma_1}\hat{c}_{\mathbf{k}_2\sigma_2}\rangle=\delta_{\mathbf{k}_1=\mathbf{k}_2}\delta_{\sigma_1=\sigma_2}f(\tilde{\epsilon}_{\mathbf{k}};\beta,\tilde{\mu}) \tag{1.11}$$

being f the Fermi-Dirac distribution,

$$f(\epsilon;\beta,\mu)\equiv\frac{1}{e^{\beta(\epsilon-\mu)}+1}$$

In other words, we are assuming a solution to exist, we call "normal", such that the net effect of the interactions is to generate an effective correction to bands such that the final hamiltonian is just $\hat{H}\simeq\hat{H}_{\tilde{t}}+(\text{shifts})$. It follows that Eq. (1.9) becomes

$$V\sum_{\langle ij\rangle}\sum_{\sigma}\left[\langle\hat{c}^\dagger_{i\sigma}\hat{c}_{j\sigma}\rangle\hat{c}^\dagger_{j\sigma}\hat{c}_{i\sigma}+\hat{c}^\dagger_{i\sigma}\hat{c}_{j\sigma}\langle\hat{c}^\dagger_{j\sigma}\hat{c}_{i\sigma}\rangle-\langle\hat{c}^\dagger_{i\sigma}\hat{c}_{j\sigma}\rangle\langle\hat{c}^\dagger_{j\sigma}\hat{c}_{i\sigma}\rangle\right]$$
$$\overset{\mathcal{F}}{=}\frac{2V}{L_xL_y}\sum_{\mathbf{K},\mathbf{k}}\sum_{\sigma}\left[\cos(2k_x)+\cos(2k_y)\right]\langle\hat{c}^\dagger_{\mathbf{K}+\mathbf{k}\sigma}\hat{c}_{\mathbf{K}+\mathbf{k}\sigma}\rangle\hat{c}^\dagger_{\mathbf{K}-\mathbf{k}\sigma}\hat{c}_{\mathbf{K}-\mathbf{k}\sigma}-E^{(\mathrm{N})}_{\mathrm{F}/V} \tag{1.12}$$

since the only contribution comes from $\mathbf{k}'=-\mathbf{k}$.

Let now $\mathbf{q}\equiv\mathbf{K}+\mathbf{k}$, $\mathbf{q}'\equiv\mathbf{K}-\mathbf{k}$. A representation of this variables change is given in Fig. 1.1. Being for $\ell=x,y$

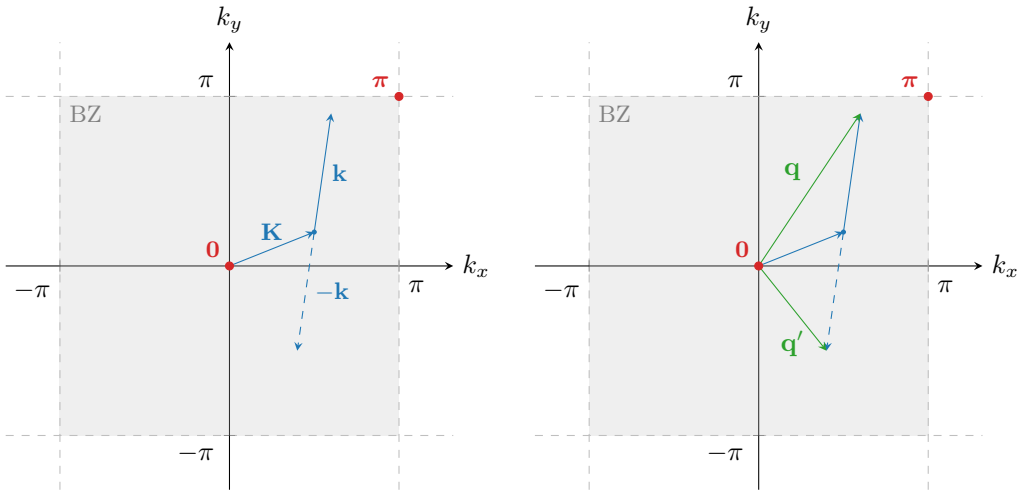
$$\begin{aligned}\delta q_\ell&\equiv q_\ell-q'_\ell\\&=(K_\ell+k_\ell)-(K_\ell-k_\ell)\\&=2k_\ell\end{aligned}$$

we reduce Eq. (1.9) to

$$V\sum_{\langle ij\rangle}\sum_{\sigma}\left[\langle\hat{c}^\dagger_{i\sigma}\hat{c}_{j\sigma}\rangle\hat{c}^\dagger_{j\sigma}\hat{c}_{i\sigma}+\hat{c}^\dagger_{i\sigma}\hat{c}_{j\sigma}\langle\hat{c}^\dagger_{j\sigma}\hat{c}_{i\sigma}\rangle-\langle\hat{c}^\dagger_{i\sigma}\hat{c}_{j\sigma}\rangle\langle\hat{c}^\dagger_{j\sigma}\hat{c}_{i\sigma}\rangle\right]$$
$$\overset{\mathcal{F}}{=}\frac{2V}{L_xL_y}\sum_{\mathbf{q},\mathbf{q}'}\sum_{\sigma}\left[\cos(\delta q_x)+\cos(\delta q_y)\right]\langle\hat{c}^\dagger_{\mathbf{q}\sigma}\hat{c}_{\mathbf{q}\sigma}\rangle\hat{c}^\dagger_{\mathbf{q}'\sigma}\hat{c}_{\mathbf{q}'\sigma}-E^{(\mathrm{N})}_{\mathrm{F}/V} \tag{1.13}$$

Recall now the result of the symmetry channels decomposition of Eq. (**??**),

$$\cos(\delta q_x)+\cos(\delta q_y)=\frac{1}{2}\sum_{\gamma}\varphi^{(\gamma)}_{\mathbf{q}}\varphi^{(\gamma)*}_{\mathbf{q}'}\quad\text{for }\gamma\in\{s^*,p_x,p_y,d_{x^2-y^2}\}$$



a | Generic vectors. b | Change of variables.

Figure 1.1 | Representation of the variable change employed to pass from Eq. (1.12) to (1.13). In Fig. 1.1a generic vectors are considered, cycling over all values of $\mathbf{K}, \mathbf{k} \in \mathrm{BZ}$. In Fig. 1.1b is depicted the variables change to the new vectors $\mathbf{q}, \mathbf{q}'$.

Thanks to this handy property, we reduce Eq. (1.13) to

$$V \sum_{\langle ij \rangle} \sum_{\sigma} \left[\langle \hat{c}^\dagger_{i\sigma} \hat{c}_{j\sigma} \rangle \hat{c}^\dagger_{j\sigma} \hat{c}_{i\sigma} + \hat{c}^\dagger_{i\sigma} \hat{c}_{j\sigma} \langle \hat{c}^\dagger_{j\sigma} \hat{c}_{i\sigma} \rangle - \langle \hat{c}^\dagger_{i\sigma} \hat{c}_{j\sigma} \rangle \langle \hat{c}^\dagger_{j\sigma} \hat{c}_{i\sigma} \rangle \right]$$
$$\overset{\mathcal{F}}{=} \sum_{\gamma,\sigma} \left[\frac{1}{L_x L_y} \sum_{\mathbf{q} \in \mathrm{BZ}} \varphi^{(\gamma)}_{\mathbf{q}} u_{\mathbf{q}\sigma} \right] \times V \sum_{\mathbf{q}' \in \mathrm{BZ}} \varphi^{(\gamma)*}_{\mathbf{q}'} \hat{c}^\dagger_{\mathbf{q}'\sigma} \hat{c}_{\mathbf{q}'\sigma} - E^{(\mathrm{N})}_{\mathrm{F}/V} \tag{1.14}$$

where we defined the EV:

$$u_{\mathbf{q}\sigma} \equiv \langle \hat{c}^\dagger_{\mathbf{q}\sigma} \hat{c}_{\mathbf{q}\sigma} \rangle \qquad u_{\mathbf{q}\sigma} = u^*_{\mathbf{q}\sigma} \tag{1.15}$$

Of course, the numerical value of $u_{\mathbf{q}\sigma}$ is given by the Fermi-Dirac distribution; however, for the sake of reproducibility of this derivation in next chapters, we omit that for a moment. Let $u^{(\gamma)}_\sigma$ be its γ-wave component,

$$u^{(\gamma)}_\sigma = \frac{1}{L_x L_y} \sum_{\mathbf{q} \in \mathrm{BZ}} \varphi^{(\gamma)}_{\mathbf{q}} \langle \hat{c}^\dagger_{\mathbf{q}\sigma} \hat{c}_{\mathbf{q}\sigma} \rangle$$

Now, the bare bands $\epsilon_{\mathbf{k}}$ are s^*-wave symmetric, but in general $\tilde{\epsilon}_{\mathbf{k}}$ can exhibit a more complex symmetry. We assume respected spatial inversion invariance, so it must be

$$u_{\mathbf{q}\sigma} = u_{-\mathbf{q}\sigma}$$

thus

$$u^{(p_\ell)}_\sigma = 0 \quad \text{for} \quad \ell \in \{x, y\}$$

recalling the antisymmetric form factors of Tab **??**. The remaining s^* and d symmetries are legitimate HFPs with relative self-consistency equations, reported in Tab. 1.2 using spin degeneracy.

The two HFPs $u^{(s^*)}_\sigma$ and $u^{(d)}_\sigma$ are in general non-zero, since we are allowing theoretically the possibility for $\mathrm{C}_4 \to \mathrm{C}_2$ SSB. Note that when imposing C_4 invariance the $u^{(d)}_\sigma$ HFP vanishes, while in general $u^{(s^*)}_\sigma$ is non-zero. From these considerations Eq. (1.14) becomes

$$V \sum_{\langle ij \rangle} \sum_{\sigma} \left[\langle \hat{c}^\dagger_{i\sigma} \hat{c}_{j\sigma} \rangle \hat{c}^\dagger_{j\sigma} \hat{c}_{i\sigma} + \hat{c}^\dagger_{i\sigma} \hat{c}_{j\sigma} \langle \hat{c}^\dagger_{j\sigma} \hat{c}_{i\sigma} \rangle - \langle \hat{c}^\dagger_{i\sigma} \hat{c}_{j\sigma} \rangle \langle \hat{c}^\dagger_{j\sigma} \hat{c}_{i\sigma} \rangle \right]$$
$$\overset{\mathcal{F}}{=} V \sum_{\mathbf{q} \in \mathrm{BZ}} \sum_{\sigma} \left[u^{(s^*)}_\sigma \varphi^{(s^*)*}_{\mathbf{q}} + u^{(d)}_\sigma \varphi^{(d)*}_{\mathbf{q}} \right] \hat{c}^\dagger_{\mathbf{q}\sigma} \hat{c}_{\mathbf{q}\sigma} - E^{(\mathrm{N})}_{\mathrm{F}/V} \tag{1.16}$$

Symmetry	Self-consistency equation	Graph
s^*-wave	$u^{(s^*)} = \frac{1}{L_x L_y} \sum_{\mathbf{k}\in\mathrm{BZ}} (\cos k_x + \cos k_y) f(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu})$	Fig. ??
$d_{x^2-y^2}$-wave	$u^{(d)} = \frac{1}{L_x L_y} \sum_{\mathbf{k}\in\mathrm{BZ}} (\cos k_x - \cos k_y) f(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu})$	Fig. ??

Table 1.2 | Symmetry resolved self-consistency equations for the harmonics of $u_{\mathbf{k}}$.

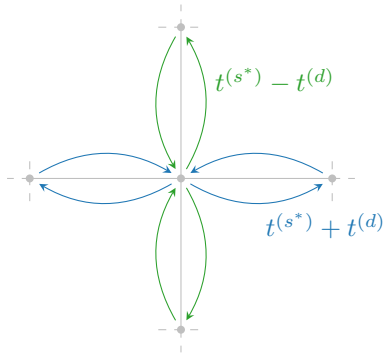


Figure 1.2 | Graphic rendition of anisotropic hopping, with a s^*-wave component (symmetric by $\pi/$ rotations) and a d-wave component (antisymmetric).

Let

$$\tilde{\epsilon}_{\mathbf{k}\sigma} \equiv \epsilon_{\mathbf{k}\sigma} + u_\sigma^{(s^*)} V(\cos k_x + \cos k_y) + u_\sigma^{(d)} V(\cos k_x - \cos k_y) \tag{1.17}$$

This equation defines the bands renormalization and resymmetrization anticipated in Tab. 1.1, and is actually spin-independent. The fact that both a s^* and a d-wave component appear introduces an interesting anisotropy that is discussed in next section.

1.3.2 Fock contraction energy

Finally, let us derive the Fock contraction energy by expanding Eq. (1.10):

$$\begin{aligned} E_{\mathrm{F}/V}^{(\mathrm{N})} &= \frac{V}{L_x L_y} \sum_{\mathbf{q},\mathbf{q}'} \sum_\sigma \left[\cos(\delta q_x) + \cos(\delta q_y)\right] \langle \hat{c}^\dagger_{\mathbf{q}\sigma} \hat{c}_{\mathbf{q}'\sigma} \rangle \langle \hat{c}^\dagger_{\mathbf{q}\sigma} \hat{c}_{\mathbf{q}'\sigma} \rangle \\ &= L_x L_y \times \frac{V}{2} \sum_{\gamma,\sigma} \left[\frac{1}{L_x L_y} \sum_{\mathbf{q}\in\mathrm{BZ}} \varphi_{\mathbf{q}}^{(\gamma)} u_{\mathbf{q}\sigma} \right] \left[\frac{1}{L_x L_y} \sum_{\mathbf{q}'\in\mathrm{BZ}} \varphi_{\mathbf{q}}^{\prime(\gamma)*} u^*_{\mathbf{q}'\sigma} \right] \\ &= L_x L_y \times \frac{V}{2} \sum_\sigma \left[\left|u_\sigma^{(s^*)}\right|^2 + \left|u_\sigma^{(d)}\right|^2 \right] \end{aligned} \tag{1.18}$$

This expression gives the Fock contraction energy coming from the $\hat{H}_V$ operator. The particularly relevant result, here, is the fact that the full spectral amplitude of $u_{\mathbf{k}\sigma}$ here is weighted and contributes to contraction energy with a V linear prefactor. This suggests that a mixed-symmetry band could be energetically preferred.

1.3.3 Nematicity: extending hopping to d-wave bands

From Tabs. ?? and ?? we can inverse-Fourier transform the form factors to get the full effective hopping,

$$\tilde{t}_{ij\sigma} = \sum_{\gamma\in\{s^*,d\}} t_\sigma^{(\gamma)} \varphi_{ij}^{(\gamma)}$$

with

$$t_\sigma^{(s^*)} = t - \frac{u_\sigma^{(s^*)} V}{2} \qquad t_\sigma^{(d)} = -\frac{u_\sigma^{(d)} V}{2}$$

Interestingly enough, in principle an anisotropy in the normal diffusion between x and y direction emerges: such a feature is referred to as "nematicity". A schematics of anisotropic mixed symmetry hopping is depicted in Fig. 1.2: as is evident, for each bond linking two sites the hopping parameter can be a positive or negative combination of the two components, whether the link direction is vertical or horizontal. Naturally, the solution needs to be invariant under exchange $x \leftrightarrow y$, making the actual SSB of the class $\mathrm{C}_4 \to \mathrm{C}_2/\mathbb{Z}_2$. For the normal phase, the physical expectation is that nematicity is suppressed; this does not make this discussion somwhat less relevant for more complex phases, such as the superconducting one, for which it is known nematicity in 2D planar lattices can arise as a source of superconducting long-range ordering [3].

The fact that hopping is here anisotropic does not break full translational invariance $\mathrm{T}_1^{(x)} \otimes \mathrm{T}_1^{(y)}$. Imposing a free degenerate Fermi gas ground state in Eq. (1.11) necessarily gives a uniform density indeed, whatever the shape of the band. By Fourier-transforming the site number operator according to the convention of Eq. (**??**),

$$\hat{c}_{\mathbf{r}\sigma}^\dagger \hat{c}_{\mathbf{r}\sigma} \overset{\mathcal{F}}{=} \frac{1}{L_x L_y} \sum_{\mathbf{k},\mathbf{k}'} e^{i(\mathbf{k}-\mathbf{k}')\cdot\mathbf{r}} \hat{c}_{\mathbf{k}\sigma}^\dagger \hat{c}_{\mathbf{k}'\sigma}$$

which obviously implies

$$\langle \hat{n}_{i\sigma} \rangle = \frac{1}{L_x L_y} \sum_{\mathbf{k}\in\mathrm{BZ}} f(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu})$$

A perfectly degenerate Fermi gas is always translational invariant and thus cannot exhibit non-uniform symmetry. The fact that hopping is allowed to be anisotropic does not interfere with that: C_4 SSB is not manifested this way.

Note that for null HFPs,

$$u_\sigma^{(s^*)} = 0 \qquad u_\sigma^{(d)} = 0$$

we have, in vector notation and recalling Tab. **??**:

$$\tilde{t}_{\mathbf{r}\mathbf{r}'\sigma} = \frac{t}{2} \left[\delta_{\mathbf{r}'=\mathbf{r}+\underline{\mathbf{x}}} + \delta_{\mathbf{r}'=\mathbf{r}-\underline{\mathbf{x}}} + \delta_{\mathbf{r}'=\mathbf{r}+\underline{\mathbf{y}}} + \delta_{\mathbf{r}'=\mathbf{r}-\underline{\mathbf{y}}} \right]$$

The fact that $t/2$ appears here instead of t is not inconsistent, it's just a consequence of the equivalence:

$$-t \sum_{\langle ij \rangle} \sum_\sigma \left[\hat{c}_{i\sigma}^\dagger \hat{c}_{j\sigma} + \mathrm{h.c.} \right] = -\frac{t}{2} \sum_{i,j\in\mathcal{S}} \sum_\sigma \left[\hat{c}_{i\sigma}^\dagger \hat{c}_{j\sigma} + \mathrm{h.c.} \right]$$

The diffusion RMP $\tilde{t}_{ij\sigma}$ is to be interpreted like the right-hand side of the above equation, with the kinetic operator becoming:

$$\hat{H}_{\tilde{t}} \equiv -\frac{1}{2} \sum_{i,j\in\mathcal{S}} \sum_\sigma \left[\tilde{t}_{ij\sigma} \hat{c}_{i\sigma}^\dagger \hat{c}_{j\sigma} + \mathrm{h.c.} \right]$$

To limit the sum to NNs is a matter of the symmetry structure of the function $\tilde{t}_{ij\sigma}$.

1.4 Normal free energy density

An important quantity to derive in order to compare different phases is free energy density. The mere existence of a MFT solution to a given phase is not enough to understand if the latter is physically established. To compare energy is instead the correct method to understand what phase is dominant. Within this section we sketch a procedure to extract free energy density we will use identically everywhere. Let

$$f_{\mathrm{MFT}}^{(\mathrm{N})} \equiv \frac{F_{\mathrm{MFT}}^{(\mathrm{N})}}{L_x L_y}$$

being F_{MFT} the total free energy. Consider the complete hamiltonian

$$\hat{H} - \mu\hat{N} \simeq \sum_{\mathbf{k},\sigma} \tilde{\epsilon}_{\mathbf{k}\sigma} \hat{c}^{\dagger}_{\mathbf{k}\sigma} \hat{c}_{\mathbf{k}\sigma} + E_{\mathrm{c}}^{(\mathrm{N})} - \tilde{\mu}\hat{N}$$

where $E_{\mathrm{c}}^{(\mathrm{N})}$ is the (negative) total contraction energy for the normal phase,

$$E_{\mathrm{c}}^{(\mathrm{N})} \equiv -\left[E_{\mathrm{H}/U}^{(\mathrm{N})} + E_{\mathrm{H}/V}^{(\mathrm{N})} + E_{\mathrm{F}/V}^{(\mathrm{N})}\right]$$

having included the various contraction energies from Eqns. (1.6), (1.7) and (1.18).

Let us now take together all terms. From now on, we will omit spin index σ, making use of the always present $\mathbb{Z}_2$ spin inversion symmetry discussed in Sec. 1.1. For a free Fermi gas, the free energy density is simply obtained by summing four contributions:

1. The free particles thermal ground state energy density,

$$e_{\mathrm{g}}^{(\mathrm{N})} = \frac{2}{L_x L_y} \sum_{\mathbf{k}\in\mathrm{BZ}} (\tilde{\epsilon}_{\mathbf{k}} - \tilde{\mu}) f(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu})$$

2. The contraction energy density,

$$\begin{aligned} e_{\mathrm{c}}^{(\mathrm{N})} &\equiv \frac{E_{\mathrm{c}}^{(\mathrm{N})}}{L_x L_y} \\ &= -n^2 U + V\left[2zn^2 - \left|u^{(s^*)}\right|^2 - \left|u^{(d)}\right|^2\right] \end{aligned}$$

3. The entropic contribution to free energy

$$e_s^{(\mathrm{N})} \equiv -Ts$$

with $T = 1/k_B\beta$ and s the free Fermi gas entropy density for a single spin sector [8],

$$s = -\frac{k_B}{L_x L_y} \sum_{\mathbf{k}\in\mathrm{BZ}} \left[(1 - f(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu})) \ln(1 - f(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu})) + f(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu}) \ln f(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu})\right] \tag{1.19}$$

We use the identity:

$$\ln\left(1 - \frac{1}{e^x + 1}\right) = x + \ln\left(\frac{1}{e^x + 1}\right)$$

and, substituting $x = \beta(\tilde{\epsilon}_{\mathbf{k}} - \tilde{\mu})$, reduce the above expression to:

$$\begin{aligned} s = -\frac{k_B}{L_x L_y} \sum_{\mathbf{k}\in\mathrm{BZ}} \Big[&\ln(1 - f(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu})) + f(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu}) \ln f(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu}) \\ &- \beta(\tilde{\epsilon}_{\mathbf{k}} - \tilde{\mu}) f(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu}) - f(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu}) \ln f(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu})\Big] \end{aligned}$$

Second and fourth terms cancel out, and multiplying by $-2T$ (taking care of spin degeneracy) we have

$$e_s^{(\mathrm{N})} = \frac{2}{\beta L_x L_y} \sum_{\mathbf{k}\in\mathrm{BZ}} \ln(1 - f(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu})) - \frac{2}{L_x L_y} \sum_{\mathbf{k}\in\mathrm{BZ}} (\tilde{\epsilon}_{\mathbf{k}} - \tilde{\mu}) f(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu})$$

4. The chemical potential correction

$$e_n^{(\mathrm{N})} = \mu n$$

Note here that we are using the "physical" value of μ, not its RMP counterpart $\tilde{\mu}$. The reason for this is discussed in Sec. **??**.

Thus, composing everything

$$\begin{aligned} f_{\mathrm{MFT}}^{(\mathrm{N})} &\equiv e_{\mathrm{g}}^{(\mathrm{N})} + e_{\mathrm{c}}^{(\mathrm{N})} + e_{s}^{(\mathrm{N})} + e_{n}^{(\mathrm{N})} \\ &= \frac{2}{\beta L_x L_y} \sum_{\mathbf{k}\in\mathrm{BZ}} \ln\left(1 - f(\tilde{\epsilon}_{\mathbf{k}};\beta,\tilde{\mu})\right) - V\left[\left|u^{(s^*)}\right|^2 + \left|u^{(d)}\right|^2\right] + \left[\mu + n(2zV - U)\right] n \end{aligned}$$

When it comes to algorithmic computation of this quantity, it is important to express μ in terms of its RMP counterpart $\tilde{\mu}$, as discussed in Sec. **??**. By doing this we are considering all the effects of MFT decomposition of $\hat{H}$. However, as was predicted, the RMP $\tilde{\mu} = \mu + n(2zV - U)$ presented in Eq. (1.5) appears because of contraction energy corrections, giving finally

$$f_{\mathrm{MFT}}^{(\mathrm{N})} = \frac{2}{\beta L_x L_y} \sum_{\mathbf{k}\in\mathrm{BZ}} \ln\left(1 - f(\tilde{\epsilon}_{\mathbf{k}};\beta,\tilde{\mu})\right) - V\left[\left|u^{(s^*)}\right|^2 + \left|u^{(d)}\right|^2\right] + \tilde{\mu} n \tag{1.20}$$

1.5 Theoretical constraints on the normal phase HFPs

Within this section we discuss some theoretical constraints on the two HFPs identified in this section. These last are useful in later numerical considerations. The discussion is hereby divided in two parts: first we consider the simpler case of unviolated C_4 symmetry, then we move to anisotropic hopping and SSB.

1.5.1 Constraints in presence of a robust C_4 symmetry and bands shrinking

Consider the simpler situation where C_4 symmetry is imposed. This implies that the renormalized bands are

$$\tilde{\epsilon}_{\mathbf{k}} \equiv (u^{(s^*)}V - 2t)(\cos k_x + \cos k_y)$$

with a single self-consistency equation

$$u^{(s^*)} = \frac{1}{L_x L_y} \sum_{\mathbf{k}\in\mathrm{BZ}} (\cos k_x + \cos k_y) f(\tilde{\epsilon}_{\mathbf{k}};\beta,\tilde{\mu})$$

Just by looking at these two equations we can conclude:

1. For $t > 0$, there is no stable solution admitting $u^{(s^*)} \leq 0$. Let us prove this: suppose instead $u^{(s^*)} \leq 0$. A generic sum over the BZ is identically written as

$$\sum_{\mathbf{k}\in\mathrm{BZ}} g_{\mathbf{k}} = \sum_{\mathbf{k}\in\mathrm{MBZ}} \left[g_{\mathbf{k}} + g_{\mathbf{k}+\boldsymbol{\pi}}\right]$$

where we defined the Magnetic Brillouin Zone (MBZ), sketched in Fig. [Missing], which will become mostly important for the AF phase and whose contour is given by the set of zeros of the s^*-wave form factor of Fig. **??**. We also defined the AF vector $\boldsymbol{\pi} \equiv (\pi,\pi)$. The self-consistency equation thus reads

$$\begin{aligned} u^{(s^*)} &= \frac{1}{L_x L_y} \sum_{\mathbf{k}\in\mathrm{MBZ}} (\cos k_x + \cos k_y)\left[f(\tilde{\epsilon}_{\mathbf{k}};\beta,\tilde{\mu}) + f(\tilde{\epsilon}_{\mathbf{k}+\boldsymbol{\pi}};\beta,\tilde{\mu})\right] \\ &= \frac{1}{L_x L_y} \sum_{\mathbf{k}\in\mathrm{MBZ}} (\cos k_x + \cos k_y)\left[f(\tilde{\epsilon}_{\mathbf{k}};\beta,\tilde{\mu}) - f(-\tilde{\epsilon}_{\mathbf{k}};\beta,\tilde{\mu})\right] \end{aligned}$$

having we used the antisymmetry of s^*-wave functions by $\boldsymbol{\pi}$ shifts. Now, if $u^{(s^*)} \leq 0$ the renormalized band $\tilde{\epsilon}_{\mathbf{k}}$ is negative inside the MBZ and positive outside, implying

$$\forall \mathbf{k} \in \mathrm{MBZ} \quad : \quad f(\tilde{\epsilon}_{\mathbf{k}};\beta,\tilde{\mu}) \geq f(-\tilde{\epsilon}_{\mathbf{k}};\beta,\tilde{\mu})$$

the equality only being satisfied on the MBZ contour. Then, since inside the MBZ it holds by definition $\cos k_x + \cos k_y \geq 0$, the above equations imply

$$u^{(s^*)} \leq 0 \quad \Longrightarrow \quad u^{(s^*)} \geq 0$$

To exclude the possibility of $u^{(s^*)} = 0$, it suffices to notice that from the argument above such a situation only arises when the MBZ contour degenerates to the entirety of the MBZ itself – which is, when the band is flat. But that cannot be the case for $u^{(s^*)} = 0$, as long as $t > 0$. Thus we have proven that

$$u^{(s^*)} > 0$$

2. For $t > 0$, there is no stable solution admitting $u^{(s^*)} \geq 2t/V$. We proceed exactly as did for the previous point, up to the passage:

$$u^{(s^*)} = \frac{1}{L_x L_y} \sum_{\mathbf{k} \in \mathrm{MBZ}} (\cos k_x + \cos k_y) \left[f(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu}) - f(-\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu}) \right]$$

Now, if $u^{(s^*)} \geq 2t/V$ this means the prefactor in front of the renormalized bands is positive. This implies the renormalized band $\tilde{\epsilon}_{\mathbf{k}}$ is positive inside the MBZ and negative outside, implying

$$\forall \mathbf{k} \in \mathrm{MBZ} \quad : \quad f(\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu}) \leq f(-\tilde{\epsilon}_{\mathbf{k}}; \beta, \tilde{\mu})$$

thus we have

$$u^{(s^*)} \geq 2t/V \quad \Longrightarrow \quad u^{(s^*)} \leq 0$$

which is absurd. Thus

$$u^{(s^*)} < 2t/V$$

Collecting the two points above, we have the constraint on $u^{(s^*)}$

$$\mathrm{C}_4 \quad \Longrightarrow \quad 0 < u^{(s^*)} < 2t/V \tag{1.21}$$

Interestingly enough, the self-consistent procedure predicts a positive convergence value for the simplest HFP, excluding the possibility of nullifying the effective hopping as well as bands sign inversion. We can denominate this net effect "bands shrinking".

1.5.2 Extension to a $\mathrm{C}_2/\mathbb{Z}_2$ symmetry

Let us extend the arguments above to a weaker symmetry, allowing for

$$\mathrm{C}_4 \overset{\mathrm{SSB}}{\to} \mathrm{C}_2/\mathbb{Z}_2$$

We work with the full, anisotropic band of Eq. (1.17). The proofs of this section are somewhat more involuted, and lead to the conclusion that within the normal phase the C_4 structure must be preserved.

Before starting, it is useful to spend a few words on the spatial structure of a mixed $s^* \otimes d$-wave band.

1. Let for instance

$$u^{(s^*)} > 2t/V > 0 \qquad u^{(d)} > 0$$

1.6 Numerical simulations

[To be continued...]

1.6.1 HF iterative algorithm sketch

We are finally able to move to the computational part of this chapter. The aim is to write down an algorithm of the kind of Sec. **??**. For the normal phase, the HFPs vector has two components:

$$\mathbf{v} = \begin{bmatrix} u^{(s^*)} \\ u^{(d)} \end{bmatrix}$$

As before, we treat spin as a pure degeneracy. The HFPs are characterized by the self-consistency equations of Tab. 1.2. The resulting model has two RMPs:

$$\tilde{t}_{ij} \equiv \left(t - \frac{u^{(s^*)}V}{2}\right)\varphi_{ij}^{(s^*)} - \frac{u^{(d)}V}{2}\varphi_{ij}^{(d)} \tag{1.22}$$

$$\tilde{\mu} \equiv \mu + n(2zV - U) \tag{1.23}$$

the second one being only dependent on the model parameters and not on HFPs. We can define the direction-dependent hopping as

$$\tilde{t}^{(x)} \equiv t - \frac{u^{(s^*)} - u^{(d)}}{2}V \tag{1.24}$$

$$\tilde{t}^{(y)} \equiv t - \frac{u^{(s^*)} + u^{(d)}}{2}V \tag{1.25}$$

Both components are controlled by V. Evidently $u^{(s^*)}$ is a measure of the direction-averaged hopping shift with respect to V,

$$u^{(s^*)} = \frac{\left(t - \tilde{t}^{(x)}\right) + \left(t - \tilde{t}^{(y)}\right)}{V}$$

while $u^{(d)}$ measures the hopping anisotropy with respect to V,

$$u^{(d)} = \frac{\tilde{t}^{(y)} - \tilde{t}^{(x)}}{V}$$

Finally, the phase free energy is given by Eq. (1.4). [To be continued...]

Bibliography

[1] James F. Annett. "Symmetry of the order parameter for high-temperature superconductivity". In: *Advances in Physics* 39.2 (Apr. 1990). Publisher: Taylor & Francis _eprint: https://doi.org/10.1080/00018739000101481, pp. 83–126. ISSN: 0001-8732. DOI: 10.1080/00018739000101481. URL: https://doi.org/10.1080/00018739000101481 (visited on 01/06/2026).

[2] Daniel P. Arovas et al. "The Hubbard Model". In: *Annual Review of Condensed Matter Physics* 13.Volume 13, 2022 (2022), pp. 239–274. ISSN: 1947-5462. DOI: https://doi.org/10.1146/annurev-conmatphys-031620-102024. URL: https://www.annualreviews.org/content/journals/10.1146/annurev-conmatphys-031620-102024.

[3] Anna E. Böhmer et al. "Nematicity and nematic fluctuations in iron-based superconductors". en. In: *Nature Physics* 18.12 (Dec. 2022), pp. 1412–1419. ISSN: 1745-2481. DOI: 10.1038/s41567-022-01833-3. URL: https://www.nature.com/articles/s41567-022-01833-3.

[4] Zhangkai Cao et al. "Dominant p-wave pairing induced by nearest-neighbor attraction in the square-lattice extended Hubbard model". In: *Phys. Rev. B* 111 (2 Jan. 2025), p. 024509. DOI: 10.1103/PhysRevB.111.024509. URL: https://link.aps.org/doi/10.1103/PhysRevB.111.024509.

[5] Piers Coleman. *Introduction to Many-Body Physics*. Cambridge University Press, 2015.

[6] Michele Fabrizio. *A Course in Quantum Many-Body Theory*. Springer, 2022.

[7] Gabriele Giuliani and Giovanni Vignale. *Quantum Theory of the Electron Liquid*. Cambridge University Press, 2005.

[8] Giuseppe Grosso and Giuseppe Pastori Parravicini. *Solid State Physics*. Second Edition. Academic Press, 2014.

[9] J. E. Hirsch. "Two-dimensional Hubbard model: Numerical simulation study". In: *Phys. Rev. B* 31 (7 Apr. 1985), pp. 4403–4419. DOI: 10.1103/PhysRevB.31.4403. URL: https://link.aps.org/doi/10.1103/PhysRevB.31.4403.

[10] Joel Hutchinson and Frank Marsiglio. "Mixed temperature-dependent order parameters in the extended Hubbard model". en. In: *Journal of Physics: Condensed Matter* 33.6 (Feb. 2021), p. 065603. ISSN: 0953-8984, 1361-648X. DOI: 10.1088/1361-648X/abc801. URL: https://iopscience.iop.org/article/10.1088/1361-648X/abc801 (visited on 01/06/2026).

[11] R. Micnas, J. Ranninger, and S. Robaszkiewicz. "Superconductivity in narrow-band systems with local nonretarded attractive interactions". In: *Reviews of Modern Physics* 62.1 (Jan. 1990). Publisher: American Physical Society, pp. 113–171. DOI: 10.1103/RevModPhys.62.113. URL: https://link.aps.org/doi/10.1103/RevModPhys.62.113 (visited on 01/06/2026).

[12] Robin Scholle et al. "Comprehensive mean-field analysis of magnetic and charge orders in the two-dimensional Hubbard model". In: *Phys. Rev. B* 108 (3 July 2023), p. 035139. DOI: 10.1103/PhysRevB.108.035139. URL: https://link.aps.org/doi/10.1103/PhysRevB.108.035139.

[13] P. Senarath Yapa et al. "Mixed-symmetry superconductivity and the energy gap". en. In: *Physica C: Superconductivity and its Applications* 633 (June 2025), p. 1354719. ISSN: 09214534. DOI: 10.1016/j.physc.2025.1354719. URL: https://linkinghub.elsevier.com/retrieve/pii/S0921453425000723 (visited on 01/06/2026).

[14] Manfred Sigrist and Kazuo Ueda. "Phenomenological theory of unconventional superconductivity". In: *Reviews of Modern Physics* 63.2 (Apr. 1991). Publisher: American Physical Society, pp. 239–311. DOI: 10.1103/RevModPhys.63.239. URL: https://link.aps.org/doi/10.1103/RevModPhys.63.239 (visited on 01/07/2026).

[15] Avinash Singh and Zlatko Tešanović. "Collective excitations in a doped antiferromagnet". en. In: *Physical Review B* 41.1 (Jan. 1990), pp. 614–631. ISSN: 0163-1829, 1095-3795. DOI: 10.1103/PhysRevB.41.614. URL: https://link.aps.org/doi/10.1103/PhysRevB.41.614 (visited on 12/12/2025).

[16] Ashvin Vishwanath. "Dirac Nodes and Quantized Thermal Hall Effect in the Mixed State of d-wave Superconductors". In: *Physical Review B* 66.6 (Aug. 2002). arXiv:cond-mat/0104213, p. 064504. ISSN: 0163-1829, 1095-3795. DOI: 10.1103/PhysRevB.66.064504. URL: http://arxiv.org/abs/cond-mat/0104213 (visited on 01/05/2026).

[17] Alexander Wietek et al. "Stripes, Antiferromagnetism, and the Pseudogap in the Doped Hubbard Model at Finite Temperature". en. In: *Physical Review X* 11.3 (July 2021), p. 031007. ISSN: 2160-3308. DOI: 10.1103/PhysRevX.11.031007. URL: https://link.aps.org/doi/10.1103/PhysRevX.11.031007 (visited on 12/12/2025).

[18] C. Zhou and H. J. Schulz. "Density of states and tunneling spectra in two-dimensional d-wave superconductors". In: *Physical Review B* 45.13 (Apr. 1992). Publisher: American Physical Society, pp. 7397–7401. DOI: 10.1103/PhysRevB.45.7397. URL: https://link.aps.org/doi/10.1103/PhysRevB.45.7397 (visited on 01/06/2026).